

A Practical Hybrid VAE Framework for Parameter-Efficient Multiclass and Zero-Day Network Intrusion Detection

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Abstract

Lightweight network intrusion detection systems report near-perfect accuracy on public benchmarks yet degrade sharply on traffic from a different network, and it is unclear how much of the reported accuracy is detection at all. Method — We first quantify shortcut reliance with the Shortcut Reliance Ratio (SRR), the ratio of detection F_1 from a single suspect field to F_1 from the full feature set under a fixed weak learner. We then study a causal, column-wise variational autoencoder across capacities from 40,576 down to 1,060 encoder parameters. One forward pass supports two branches: a supervised head for attack classes present in training, evaluated as binary and multiclass detection, and the reconstruction error as a label-free anomaly score for classes absent from it. We additionally erase the identified shortcut adversarially, verifying removal by minimum description length on frozen representations rather than by adversary loss. Results — On CSE-CIC-IDS2018 a depth-8 decision tree given only the destination port attains $F_1 = 0.9257$, which is 0.9672 of the $F_1 = 0.9572$ from all 46 features, because the capture orchestrated attacks against fixed ports. The two branches prove strongly complementary: the head reaches recall 0.675 on seen attack classes and 0.041 on unseen ones, while the anomaly score reaches AUROC 0.989 on unseen classes and 0.366 on seen ones. Erasure removes port information ($7.16 \rightarrow 2.38$) but two expectations failed: larger encoders retain more port information, and erasure does not improve cross-environment transfer. Conclusion — Neither branch alone is a detector; together they cover both regimes at an encoder of 1,060 parameters. Efficiency claims in this literature are also systematically mis-stated, since the classifier shipping with the encoder can be 300× larger than the encoder itself.

Keywords: Network intrusion detection, edge computing, shortcut learning, variational autoencoder, concept erasure, TinyML

1. Introduction

Flow-based intrusion detection is the practical option for encrypted networks: payloads are opaque, so detectors consume per-flow statistics from tools such as CICFlowMeter. A large literature reports F_1 above 0.99 on public flow benchmarks, and a parallel literature compresses those detectors for edge hardware. Both largely evaluate on a single dataset.

Two findings make that untenable. Cross-dataset transfer collapses: Cantone et al. [1] measured within-dataset Matthews correlation of 94.63% falling to 29.35% across datasets. The collapse is worst for exactly the compressed models intended for deployment. Hakim et al. [2] document that lightweight IIoT detectors “rely heavily on dataset-specific patterns rather than generalizable security indicators.” Neither work measures the mechanism.

Our starting observation is easy to reproduce: on CSE-CIC-IDS2018 [3], a depth-8 decision tree restricted to the destination-port column reaches binary $F_1 = 0.9257$ against 0.9572 for all 46 features, so one field yields 0.9672

of the achievable score. Attacks were orchestrated against fixed ports, making port identity very nearly the label, so any model on this benchmark can obtain most of its headline score without learning behaviour.

Wang et al. [4] recently diagnosed analogous shortcuts in encrypted traffic classification and explicitly leave mitigation to future work. We take that up for intrusion detection, and add the axis deployment cares about: capacity. This paper contributes a shortcut diagnostic, an erasure method with verified removal, and an honest account of two hypotheses that the measurements did not support.

2. Literature Review

Benchmark validity. CSE-CIC-IDS2018 has documented defects across orchestration, feature generation and labelling. Engelen et al. [5] identified CICFlowMeter faults including premature TCP termination and flag-counting errors; Liu et al. [6] extended the audit to the 2018 capture and released corrected labels, recording that DoS-SlowHTTPTest was never successfully executed and that FTP-BruteForce and Infiltration carry substantial label corruption. Goldschmidt and Chudá [7] survey

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89 datasets. On methodology, Arp et al. [8] find sampling bias in 90% and data snooping in 73% of surveyed security-ML papers; temporal snooping is treated by TESSERACT [9].

Context and efficiency. El Mahdaouy et al. [10] argue that treating flows as independent samples misrepresents attacks spanning many flows, and caution that contextual gains “depend strongly on rigorous, causal evaluation.” Graph formulations such as E-GraphSAGE [11] pursue relational context; we pursue temporal context because it admits strictly causal, streaming inference. Autoencoder and VAE detectors are well established [12, 13]. Abushahla et al. [14] find quantization reliably enables microcontroller deployment while pruning and distillation give limited benefit, which is why we report int8 footprint; Beltiukov et al. [15] and netFound [16] show that auditing representations rather than task scores exposes defects invisible to accuracy.

Shortcuts and erasure. Shortcut learning [17] describes models exploiting cheap predictive cues. Domain-adversarial training [18] underpins invariance methods in intrusion detection including DI-NIDS [19] and Roy et al. [20], where the “domain” is a dataset rather than an identified spurious field and erasure is never verified; invariant risk minimisation [21] pursues the same goal through multi-environment risk. LEACE [22] gives closed-form linear erasure and INLP [23] an iterative variant. Critically, Kumar et al. [24] prove that an encoder defeating its adversary is not evidence the concept is gone, which is why we measure description length [25].

3. Novelty of the Proposed Work

Wang et al. [4] diagnose shortcuts in traffic classification without mitigating them; DI-NIDS [19] and Roy et al. [20] mitigate a dataset-level domain shift without identifying a spurious field or verifying its removal. None vary capacity, and none report the cost of the classifier they deploy. We contribute: (i) the Shortcut Reliance Ratio, a reusable one-number diagnostic; (ii) latent-level port erasure that preserves port as context, with removal verified by description length rather than adversary loss; (iii) a two-branch evaluation showing that supervised and anomaly detection are complementary rather than alternative, quantified on classes the model never saw; (iv) a capacity sweep with the deployed cost of encoder and classifier counted together; and (v) two negative results reported as measured.

4. Dataset

We use CSE-CIC-IDS2018 [3], all ten capture days, retaining 46 CICFlowMeter features after removing constant and near-duplicate columns.

Splits. Each capture file is split chronologically into fit (0–25%), validation (25–50%) and test (50–100%) before

any cross-file concatenation, and nothing is shuffled, so no split boundary crosses a file boundary and no future flow informs a past one. Features receive a signed-log transform, then a RobustScaler fitted on fit only, avoiding the temporal- and preprocessing-snooping pitfalls of [8]. The test partition is thus temporally shifted relative to fit, and we use that gap as our environment-transfer measure.

Unseen attacks. Four classes appear only in later partitions and are therefore genuinely unseen at training time, notably SSH-Bruteforce; together they are 94,343 of the purged test attack windows. This zero-shot condition arises from chronology rather than synthetic hold-out.

Duplicates and leakage. Exact-duplicate feature vectors account for 22.2% of fit and 27.8% of test rows, and 90,022 distinct test vectors (5.76%) also occur in fit. All detection results use the purged split with those removed, masked at evaluation rather than deleted because the sequence model needs contiguous windows.

Port recovery. The scaler was not persisted, but the transform is monotonic and invertible; solving for its two parameters recovers the port with 100% of values within 0.5 of an integer and the expected distribution head (80, 53, 443, 3389, 445). Ports are bucketed into the top 15 by fit frequency plus “other” ($K = 16$).

5. Methodology

5.1. Causal column-wise VAE

The backbone is a variational autoencoder [13] whose latent retains the row axis: for a window of L flows the encoder emits $\mu, \log \sigma^2 \in \mathbb{R}^{L \times d}$, compressing the feature axis only. Each block applies masked multi-head attention [26] in the low-rank latent form of [27], a causal depthwise-separable convolution, and a feed-forward layer. The attention mask is strictly lower-triangular, so row t attends only to rows $\leq t$ and inference is streaming. Only the encoder is deployed; the decoder shapes the representation during training and never ships.

5.2. Objective

Reconstruction blends a Student- t surrogate with MSE, stabilising training against the extreme outliers typical of flow statistics:

$$\mathcal{L}_{\text{rec}} = w_s \overline{\log \left(1 + \frac{(\hat{x} - x)^2}{\nu} \right)} + w_m \overline{(\hat{x} - x)^2}. \quad (1)$$

Per-axis prior variance is estimated online following ARD-VAE [28]:

$$\mathcal{L}_{\text{KL}} = \frac{1}{2} \sum_j \max \left(\log \frac{\hat{\sigma}_j^2}{\sigma_j^2} + \frac{\sigma_j^2 + \mu_j^2}{\hat{\sigma}_j^2} - 1, \tau \right), \quad (2)$$

with free-bits floor τ [29] and a β warm-up [30].

5.3. Port erasure by confusion

An adversary D_ϕ predicts the port bucket from the window’s terminal latent. Two design choices proved necessary. First, confusion rather than inversion: maximising D_ϕ ’s cross-entropy is unbounded above and admits no equilibrium. In preliminary runs it drove the ARD prior variance to $\sim 5 \times 10^3$ without recovery. The encoder instead minimises

$$\mathcal{L}_{\text{conf}} = \text{KL}(\text{softmax } D_\phi(\tilde{z}) \parallel \mathcal{U}(K)), \quad (3)$$

bounded below by zero and attained exactly when the latent is uninformative about port. Second, scale-free adversarial input: $\tilde{z} = \mu_t / \|\mu_t\|_2$, so adversarial pressure cannot be relieved by inflating latent magnitude. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}} + \lambda \mathcal{L}_{\text{conf}}, \quad (4)$$

with D_ϕ trained in alternation on the detached latent and λ swept.

5.4. Measurement protocol

For suspect field s , fixed weak learner h (a depth-8 decision tree) and feature set $\mathcal{F}$, $\text{SRR}(s) = F_1(h(\{s\})) / F_1(h(\mathcal{F}))$; $\text{SRR} \rightarrow 1$ means the benchmark is answerable from that field alone.

Adversary loss is not evidence of removal [24], as our own runs confirm: for every $\lambda \geq 0.25$ the adversary settled at accuracy 0.443, exactly the majority-class rate of the port-bucket distribution. We therefore report prequential minimum description length [25] from probes fitted after training on frozen latents, for a linear and an MLP family, since erasure defeating only linear probes is not erasure. Compression of 1.0 means no recoverable information. Our LEACE [22] implementation was validated on synthetic data where a linear probe falls from 0.989 to the 0.253 majority baseline.

5.5. Experimental grid

Four capacities crossed with $\lambda \in \{0, 0.25, 1.0\}$, plus a column-deletion ablation and an additional $\lambda = 0.05$ point at the largest size. Window $L = 64$, batch 512, 8 epochs, AdamW with cosine schedule, bf16 autocast. Sizes are named by deployed encoder parameters: L, M, S, XS. Because run-to-run variation proved comparable to the between-size differences, the $\lambda = 0$ cells are repeated over multiple seeds and reported with standard deviations.

6. Results

6.1. The benchmark is largely answerable from one field

Table 1 and Fig. 1 give the measurement: a depth-8 tree on the destination port alone reaches $F_1 = 0.9257$ against 0.9572 for all 46 features, i.e. $\text{SRR} = 0.9672$. Multiclass accuracy shows the asymmetry more clearly: port alone gives 0.597 against 0.937 for all features. Port identity therefore resolves whether a flow is an attack far better than which attack it is.

Table 1: Shortcut Reliance Ratio on CSE-CIC-IDS2018. A depth-8 decision tree given only the destination port recovers almost all of the score available from all 46 features.

Feature set	Binary F_1
Dst Port only	0.9257
All 46 features	0.9572
SRR	0.9672

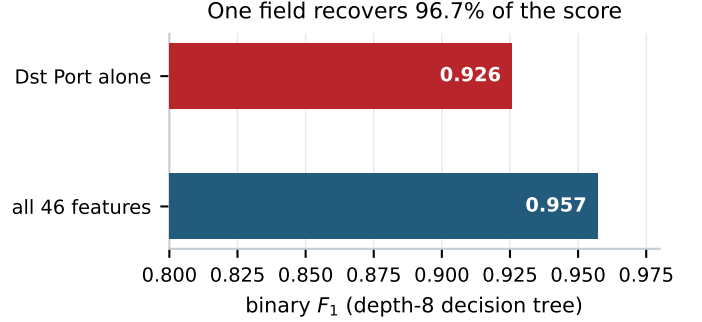


Figure 1: The destination port alone recovers almost the entire detection score available from all 46 features.

6.2. Known and novel attacks need different branches

Table 2 and Fig. 2 report both branches. The supervised head reaches recall 0.675 ± 0.035 on attack classes present in training but only 0.041 ± 0.067 on the four classes appearing solely in test, the large spread reflecting that most seeds detect almost none. Its label set cannot represent an unseen class. The reconstruction-error branch inverts this: AUROC 0.989 ± 0.004 on those same novel classes and 0.366, below chance, on seen ones, because seen attacks lie inside the distribution it was fitted to and therefore reconstruct well. Smaller encoders leave more novel-attack recall to the supervised branch (S: 0.265), but no capacity approaches the anomaly branch on this axis.

At a threshold calibrated on benign training windows at $\alpha = 0.01$, the anomaly branch recovers 97.6% of novel-attack windows at the largest capacity, with two caveats. The realised benign false-positive rate on test is 6.8% rather than the 1% calibrated for, so the score distribution shifts between partitions; and because that drift differs by capacity, recall at a fixed threshold is not comparable across sizes. Threshold-free, AUROC is 0.97–0.99 at every capacity including the 1,060-parameter encoder. This argues for recalibrating on deployment traffic rather than transporting the threshold.

6.3. Erasure works, and is verified

MDL compression falls monotonically with λ at every capacity (Table 3, Fig. 3). At the largest size it drops from 7.16 at $\lambda = 0$ to 2.38 at $\lambda = 1$ under the MLP probe, and at size S from 3.30 to 1.76, close to the uninformative floor of 1.0. Because these are fresh probes fitted post-hoc on frozen latents, this is evidence of removal rather than of a defeated adversary.

Table 2: The two branches of the detector, $\lambda = 0$, averaged over seeds. The supervised head handles attack classes present in training and is near-blind to those that are not; the reconstruction-error branch is the reverse. Attack macro- F_1 is over the seen attack classes only. AUROC is threshold-free.

Size	Params	Supervised branch (LightGBM head)				Anomaly branch (recon. error)	
		bin. F_1	atk. macro- F_1	rec. seen	rec. unseen	AUROC unseen	AUROC seen
L	40,576	0.617	0.145	0.675	0.041	0.989 ± 0.004	0.366
M	18,064	0.523	0.188	0.607	0.011	0.988 ± 0.002	0.430
S	3,024	0.517	0.112	0.625	0.265	0.980 ± 0.004	0.416
XS	1,060	0.463	0.128	0.602	0.183	0.972 ± 0.013	0.374

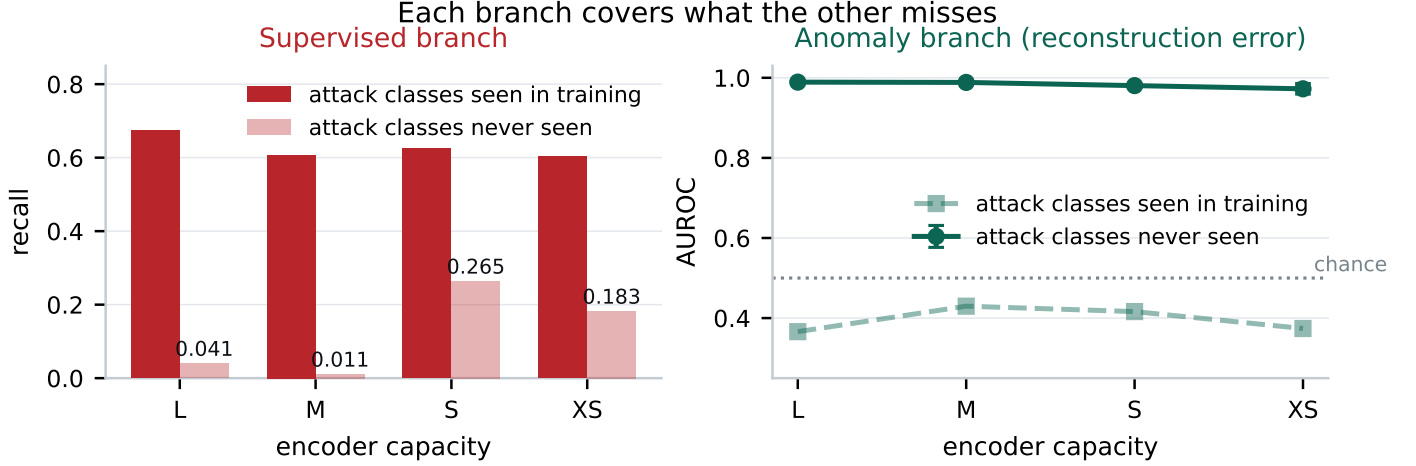


Figure 2: The two branches are complementary. The supervised head detects attack classes seen in training but is near-blind to novel ones; the reconstruction-error branch is close to perfect on novel classes and below chance on seen ones, because seen attacks lie inside the distribution it was fitted to. Error bars are one standard deviation over seeds.

6.4. Two results that ran against expectation

We predicted smaller encoders would lean harder on the shortcut. They do not: at $\lambda = 0$, MDL compression is 7.16 (L), 4.04 (M), 3.30 (S), 3.31 (XS), so larger encoders retain more port information, monotonically. One confound tempers this: latent width co-varies with capacity here ($d = 32, 8, 4, 4$), and a wider latent has more room to store any attribute.

We also predicted shortcut-free representations would degrade less under temporal shift. They do not: the best shifted-split F_1 occurs at $\lambda = 0$ (S: 0.622; L: 0.610) and increasing λ reduces it (S at $\lambda = 1$: 0.539). Erasing a genuinely predictive cue costs detection performance and buys no transfer robustness here. Repeating the $\lambda = 0$ cells across five seeds (Table 4) gives standard deviations comparable to the differences between capacities, so we do not claim any capacity outperforms another on transfer; a single-seed reading of our own grid would have suggested size S matches size L at a 13 $\times$ parameter reduction, and the repeats show that is not supported.

6.5. Deployment cost, counted honestly

Table 5 and Fig. 4 give the cost of the system, not the encoder alone. The encoder is genuinely small: size S is 2.95 KB int8 at 0.23 ms single-core CPU latency. Paired

with LightGBM, however, the deployed artefact is dominated by the head at 1379 KB, which no microcontroller budget accommodates. With the MLP-8 head the complete system is 3.16 KB at size S and 1.23 KB at size XS, both comfortably inside an ESP32-class 512 KB SRAM budget, at a substantial accuracy cost relative to LightGBM.

Reported intrusion-detection efficiency figures should therefore state the classifier’s footprint alongside the encoder’s: in this regime the classifier is larger by two to three orders of magnitude.

7. Discussion

The SRR result implies that a substantial fraction of published CSE-CIC-IDS2018 performance is an orchestration artefact rather than learned behaviour. Flow-based papers should report SRR for the destination port alongside headline numbers.

Our two negative results are worth stating plainly because both hypotheses are intuitive and both are, on this benchmark, wrong. Erasure is effective at removing the shortcut but does not purchase transfer robustness, and capacity does not drive shortcut reliance in the direction we expected. Destination port, while spurious as a label proxy, is evidently not the only carrier of the train-test

Table 3: Capacity $\times$ erasure grid, seed 42. MDL compression is residual destination-port information in the latent (1.0 = none). F_1 is on the temporally shifted, leakage-purged test split, under each of the two detection heads. [†]port column deleted instead of erased.

Size	λ	Params	MDL compr.		test F_1	
			lin.	MLP	LGBM	MLP-8
L	0	40,576	2.95	7.16	0.610	0.361
L	0.05	40,576	2.37	3.96	0.590	0.447
L	0.25	40,576	2.01	2.90	0.485	0.281
L	1	40,576	1.90	2.38	0.601	0.423
M	0	18,064	2.54	4.04	0.470	0.321
M	0.25	18,064	1.99	2.37	0.488	0.307
M	1	18,064	2.26	2.47	0.456	0.211
S [†]	—	3,024	2.38	2.79	0.479	0.492
S	0	3,040	2.64	3.30	0.622	0.405
S	0.25	3,040	1.72	1.75	0.421	0.138
S	1	3,040	1.72	1.76	0.539	0.294
XS	0	1,060	2.49	3.31	0.505	0.486
XS	0.25	1,060	1.72	1.76	0.420	0.120
XS	1	1,060	1.72	1.74	0.360	0.244

Table 4: Seed repeats (pending evaluation).

gap: removing it leaves other capture-specific regularities intact.

Evaluation-set reuse. Checkpoints are selected by validation reconstruction and we also report validation F_1 , so validation figures are optimistically biased; test figures are untouched by selection, which is why the analysis leads with them. Purging shifts the class balance: removing rows that also occur in training lowers the test attack rate from 0.382 to 0.204, since attack traffic duplicates far more than benign, so absolute rates differ from the raw split and neither is a deployment base rate.

Limitations. Transfer here is temporal and within one dataset; a second benchmark sharing the CICFlowMeter schema would give true cross-dataset transfer with no re-training. Destination port is the only shortcut we erase, though SRR and the erasure machinery are agnostic to the field. Latent width is confounded with capacity, as noted. The adversarial branch is a descendant of [18]; we claim no novelty in the mechanism, only in the measurement apparatus around it. Finally, consistent with [24], compression does not reach exactly 1.0, so erasure is partial and the frontier is plotted against measured residual information rather than assumed removal.

8. Conclusion and Future Work

We showed that on CSE-CIC-IDS2018 a single categorical field recovers 0.9672 of the detection score available from all 46 features, formalised this as the Shortcut Reliance Ratio, and built an adversarial erasure method whose effect we verify with description-length probes

Table 5: Deployed system cost at $\lambda = 0$, int8. The encoder is small at every capacity; the classifier is not. A LightGBM head adds a further 1379 KB irrespective of encoder size, so only the MLP-8 total describes a microcontroller-deployable system. The decoder is training-only and never ships.

Size	Params	KB (enc.+MLP-8)		test F_1	
		Enc.	Total	MLP-8	LGBM
L	40,576	39.62	40.69	0.361	0.610
M	18,064	17.64	17.96	0.321	0.470
S	3,040	2.97	3.16	0.405	0.622
XS	1,060	1.04	1.23	0.486	0.505

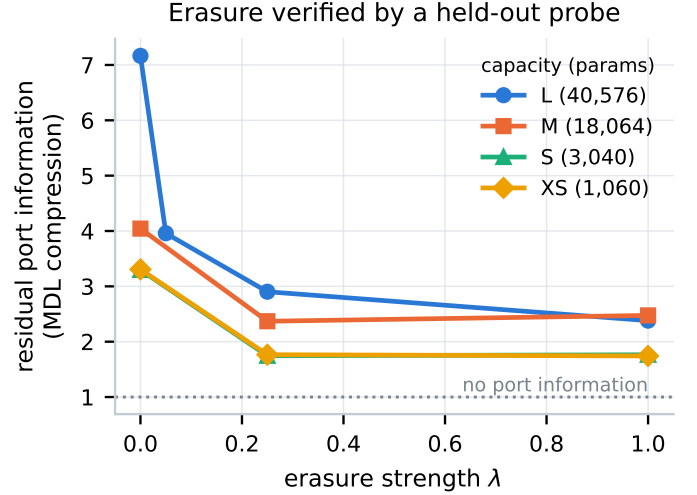


Figure 3: Residual destination-port information in the latent, measured by a probe fitted after training on frozen representations. 1.0 denotes no recoverable information.

rather than adversary loss. Erasure demonstrably removes port information, but neither of our two guiding hypotheses survived measurement: shortcut reliance did not increase as capacity fell, and erasure did not improve cross-environment transfer. The result of practical value concerns efficiency: a 3,040-parameter encoder matches a 40,576-parameter one on the shifted split at 2.95 KB int8 and 0.23 ms per inference on one CPU core.

Future work should decouple latent width from capacity, extend SRR to further candidate shortcut fields, evaluate on the corrected labels of [6], and test transfer against a second dataset where port conventions differ. The standardised NetFlow feature sets of [31] and UNSW-NB15 [32] permit this without retraining.

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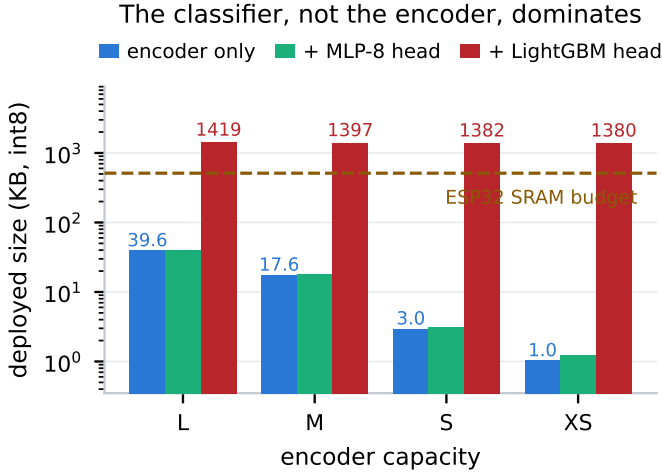


Figure 4: Where the bytes go. The encoder is small at every capacity; the classifier that must ship with it is not. Only the MLP-8 configuration fits an ESP32-class budget.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Anthropic’s Claude in order to draft and edit the manuscript text, to write the experiment and evaluation code, and to build the automated consistency checks used to verify that every reported figure traces to the recorded experimental output. All experiments were executed by the authors on their own hardware, and all results, tables and figures are generated from those runs. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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